

Chapter 3: The Plant Cell

3.1 Cell Wall

The plant cell wall is composed of cellulose microfibrils. It provides structural support and protection.

The cell membrane lies just inside the cell wall. It is selectively permeable, controlling what enters and exits.

The cytoplasm contains various organelles including:

- Chloroplasts (photosynthesis)
- Mitochondria (cellular respiration)
- Ribosomes (protein synthesis)
- Endoplasmic reticulum (transport)

Fig. 3.1: Plant Cell

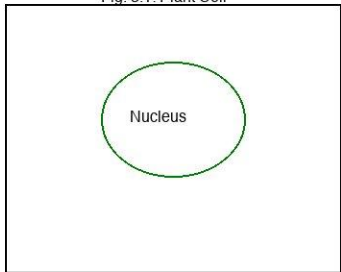


Fig. 3.1: A typical plant cell showing major organelles.

3.2 Ribosomes and Endoplasmic Reticulum

Ribosomes are the sites of protein synthesis. They can be found free in the cytoplasm or attached to the endoplasmic reticulum.

Ribosomes translate mRNA into polypeptide chains.

The rough endoplasmic reticulum (RER) has ribosomes on its surface and is involved in protein processing and transport.

The smooth endoplasmic reticulum (SER) lacks ribosomes and is involved in lipid synthesis and detoxification.

Free ribosomes produce proteins for use within the cell.

Bound ribosomes produce proteins for export or membrane use.

Each ribosome consists of two subunits: a large and a small.

3.3 Mitochondria

Mitochondria are the powerhouse of the cell. They perform cellular respiration, converting glucose and oxygen into ATP (adenosine triphosphate).

Each mitochondrion has a double membrane:

- Outer membrane: smooth, permeable to small molecules
- Inner membrane: folded into cristae, site of ATP synthesis
- Matrix: contains enzymes for the Krebs cycle
- Intermembrane space: between the two membranes

The number of mitochondria varies by cell type. Muscle cells have many mitochondria due to high energy demands.

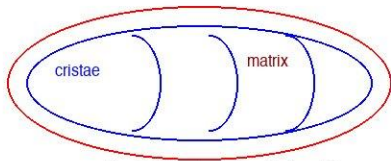


Fig. 3.2: Structure of a mitochondrion